Project Framework Discussion

Working Title: VR-Guided Shared Teleoperation Framework for Humanoid Robots

1. Motivation

Humanoid robots are expected to operate in hazardous, remote, and human-centered environments where intuitive teleoperation is essential. Existing solutions mainly rely on either full-body teleoperation, which imposes a high cognitive burden on the operator, or fully autonomous systems, which reduce human involvement but limit flexibility. This project explores a shared teleoperation paradigm that combines human intention with autonomous whole-body control.

2. Core Idea

The key concept is to separate task specification from motion execution. The human operator specifies what the robot should achieve through immersive VR interaction, speech, and gestures, while the humanoid robot autonomously determines how to achieve the desired target safely and stably. Instead of directly controlling every joint, the operator continuously provides target poses and high-level commands.

3. Overall Framework

Human Operator
↓
VR Head & Hand Tracking + Gesture + Speech
↓
Human Intention Layer
↓
Target Pose Manager
↓
Vision Perception + Robot State
↓
Autonomous Whole-body Planner
↓
Whole-body Controller
↓
Humanoid Robot
↓
VR Visual Feedback

4. Functional Modules

Module 1 – Human Interaction
• Acquire head and hand poses from VR devices.
• Recognize speech commands.
• Recognize gesture commands.

Module 2 – Target Pose Manager (Core Module)
• Fuse continuous VR poses with discrete commands.
• Generate unified target poses and task objectives.
• Manage robot operation modes.

Module 3 – Environment Perception
• Object detection.
• Scene understanding.
• Traversability analysis.
• Reachability analysis.
• Obstacle detection.

Module 4 – Autonomous Whole-body Motion
• Whole-body motion planning.
• Footstep generation.
• Balance control.
• Collision avoidance.
• Whole-body coordination.

Module 5 – VR Feedback
• Robot first-person view.
• Robot status visualization.
• Task feedback.

5. Information Flow

Operator Input
↓
Target Generation
↓
Environment Understanding
↓
Motion Generation
↓
Robot Execution
↓
Visual Feedback
└──────→ Operator

6. Initial Research Questions

RQ1. How can VR poses, speech, and gestures be fused into a unified target representation?

RQ2. How can the robot autonomously generate dynamically feasible whole-body motions from user-defined target poses?

RQ3. How can visual perception improve motion planning, stability, and collision avoidance?

RQ4. How can operator workload be reduced while maintaining intuitive control and high task performance?

7. Expected Contributions (Initial)

• A hierarchical shared teleoperation framework for humanoid robots.
• A Target Pose Manager for multimodal human inputs.
• Vision-guided autonomous whole-body motion generation.
• An immersive VR-based interaction framework integrating teleoperation and autonomous control.
• A foundation for future research on humanoid shared autonomy.

8. Discussion Points

This framework represents the initial concept for discussion rather than a finalized algorithm. Future work will focus on refining the Target Pose Manager, investigating whole-body planning strategies, selecting appropriate control algorithms, and defining quantitative evaluation metrics. The primary objective at this stage is to establish a clear system architecture and identify the core scientific questions that will guide subsequent research.